MFT (Multiple Choice Questions)

1. A PDA chooses the next move based on [BL-1]

   a) Current state

   b) Next input symbol

   c) Both (a) and (b)

   d) None of these

2. Which type of symbols contain in the stack of PDA [BL-1]

   a) Variable

   b) Terminal

   c) Both (a) and (b)

   d) None of these

3. A push down automata can be represented using: [BL-1]

   a) Transition graph

   b) Transition table

   c) ID

   d) All of the mentioned

4. A language is accepted by a push down automata if it is: [BL-2]

   a) regular

   b) context free

   c) both a and b

   d) none of the mentioned

5. The following move of PDA is on the basis of: [BL-2]

   a) Present space

   b) input symbol

   c) both a and b

   d) none of the mentioned

6. A string is accepted by a PDA when... [BL-1]

   a) Stack is empty

   b) acceptance state

   c) both a and b

   d) none of the mentioned

7. Which of the operation are eligible in PDA [BL-1]

   a) push/pop

   b) delete

   c) insert

   d) none of these

8. Push down automata accepts _______ languages [BL-1]

   a) Type 3

   b) Type 2

   c) Type 1

   d) Type 0

9. A push down automaton employs ______ data structure [BL-1]

   a) Queue

   b) Linked List

   c) Hash Table

   d) Stack

10. Which among the following is a CFG for the given Language

    L={x∈{0,1}* | number of zeros in x = number of one's in x} [BL-2]

    a) $S \rightarrow 0S1|1S0|SS|\varepsilon$

    b) $S \rightarrow 0B|1A|A \rightarrow 0S\ B \rightarrow 1S$

    c) All of the mentioned

    d) none of these`

Long Answer

11. Give leftmost and rightmost derivations of the following strings:
    a) 00101
    b) 1001
    c) 00011 [BL-3]
12. Construct PDA which accepts the language $L = \{a^n cw^n \mid w \in \{a, b\}, c \in \Sigma\}$ [BL-3]
13. Construct PDA for the language $L = \{a^n b^n \mid a, b, \in \{a, b\}, n \geq 0\}$ [BL-3]
14. Design a PDA to accept the set of all strings of 0's and 1's such that no prefix has more 1's than 0's [BL-3]
15. Construct PDA: Accepting the set of all strings over {a, b} with equal number of a's and b's. Show the moves for **abbaba** [BL-3]
16. Construct PDA to accept by final state the language of all strings of 0's and 1's such that number of 1's is less than number of 0's. Also convert the PDA to accept by empty stack [BL-3]
17. Enlist the steps to convert PDA to CFG with suitable example [BL-2]
18. Explain push down automata with an example [BL-2]
19. Consider the following CFG: G = ({S, A}, (a, b), P, S) Where P consists of:
    $S \rightarrow aAs \mid a$
    $A \rightarrow SbA \mid ss \mid ba$
    Derive string 'aabbaa' using leftmost & rightmost derivation
20. Convert given CFG into GNF [BL-3]:
    $S \rightarrow Bs \mid Aa$

$A \rightarrow Bc$

$B \rightarrow Ac$ where, $V = \{S, A, B\}$ & $T = \{a, c\}$

21. Eliminate the ε-productions from the Grammar G which is defined as: [BL-3]

$S \rightarrow ABA$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid \varepsilon$

22. Design CFG for the following Languages [BL-3]

i) $L = \{a^i b^j c^k \mid i = j + k\}$ ii) $L = \{a^{2n} b^n c^n \mid n \geq 1\}$/

ii) L = {a2n bc/ n≥ 1}

23. Consider CFG with productions [BL-3]

$S \rightarrow aB/bA$

$A \rightarrow a/aS/bAA$

$B \rightarrow b/bS/aBB$

For string **"aaabbabbba"** find leftmost and rightmost derivation.

24. Convert the given regular grammar to its equivalent FA [BL-3]

$S \rightarrow aS \,/\, bS \,/\, aA$

$A \rightarrow bB$

$B \rightarrow aC$

$C \rightarrow a$

25. Define CNF and GNF with suitable examples [BL-2]